Prajwal Singh

7398314111 | prajwalsingh0605@gmail.com | LinkedIn | Github

Education

MVJ College of Engineering — Bengaluru, Karnataka
B.E. in Information Science and Engineering (CGPA: 8.29) — *Nov 2022 – Jun 2026*

St. Mary's Convent School — Varanasi, Uttar Pradesh
Intermediate: 79.2% | Matriculation: 88.8% — *May 2019 – July 2021*

Experience

Option Circle — Bengaluru, Karnataka
Backend Developer · Registered as Exaultant Trading Circle — *Jun 2026 – Present*

- Analyze and resolve production bugs in a real-time algorithmic options trading platform, debugging live order-execution, market-data, and position-management flows across distributed microservices.
- Implemented a dynamic trailing stop-loss engine in Redis Lua (EVAL/EVALSHA): atomic per-tick read–modify–write cycles track peak profit and derive dynamic exit thresholds, eliminating race conditions between concurrent market ticks.
- Designed profit-trigger activation with trailing state persisted in Redis, publishing close events via Redis Pub/Sub to decouple exit detection from order-closing execution.
- Evaluated migrating the market-data pipeline (Polygon WebSocket → Kafka → Telegraf → InfluxDB) to Redis Streams, designing a Go-based bridge for Telegraf interoperability to reduce infrastructure complexity.

Option Circle — Bengaluru, Karnataka
Software Engineering Intern – Backend · Registered as Exaultant Trading Circle — *Jan 2026 – May 2026*

- Engineered a high-performance backend microservice (`conversion-service`) using Java 11 and Spring Boot to translate complex UI workflows into canonical schemas for automated options trading.
- Architected a dynamic schema mapping pipeline using LuaJ, enabling the execution of 2,700+ lines of Lua transformation scripts for payload generation without backend recompilation.
- Authored 900+ lines of robust Java-based validation logic to enforce data integrity constraints and achieve functional parity with UI-level validation.
- Integrated the Gemini AI API via Spring WebFlux clients to optimize natural language prompt generation for volatility trading agents and technical gating decisions.

Projects

TickPulse – Real-Time Market Alert & Trailing-Stop Engine — *Jul 2026*

- *Java 17, Spring Boot, Redis Streams, Lua, Redis Pub/Sub, PostgreSQL, Docker, WebSocket, SSE, Testcontainers*
- Built an event-driven engine ingesting live crypto ticks from the Binance WebSocket into Redis Streams, processed by consumer groups with at-least-once delivery, XACK acknowledgment, and pending-list recovery on restart.
- Implemented atomic trailing-stop and price-alert evaluation as Redis Lua scripts (EVALSHA) with peak-profit tracking and dynamic exit thresholds; verified the full state machine with Testcontainers integration tests.
- Streamed triggered alerts to clients over Server-Sent Events via Redis Pub/Sub fan-out; persisted alert history in PostgreSQL and exposed validated REST APIs with Actuator health and metrics.

Cooked – AI-Powered Backend Aggregation System — *Sep 2025 – Dec 2025*

- *Java, Spring Boot, REST APIs, OAuth 2.0, Gemini API*
- Developed a backend service aggregating user data from Spotify, GitHub, and Reddit via secure OAuth flows.
- Integrated Gemini API using Java `HttpClient` for AI-based structured persona generation.
- Handled API errors and implemented rate-limit safeguards for stable third-party integration.

Technical Skills

Languages: Java, SQL, Lua, Python, C, C++

Backend: Spring Boot, Spring WebFlux, Spring Data JPA, REST APIs, Microservices

Data & Messaging: Redis (Streams, Pub/Sub, Lua scripting), Elasticsearch, Cassandra, MySQL, PostgreSQL, Kafka, InfluxDB

Tools: Git, GitHub, Docker, Postman, JMeter, IntelliJ IDEA

Concepts: Event-Driven Architecture, Concurrency, Caching Strategies, OAuth 2.0, Generative AI (LLMs), Data Structures & Algorithms